



Palo Alto Networks Cybersecurity Academy – Setting up Web Filtering for an Elementary School Environment

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Purpose

This lab helps to develop foundational cybersecurity and management skills regarding URL filtering. This lab develops understanding of network management and configuration so that users on the network can be productive and safeguarded from malicious sites. Concepts such as certificates, TLS/SSL, and policy implementation are reinforced in this lab.

Background

Google reports 9,500 malicious websites everyday now totaling over 5,000,000 malicious websites that are out to steal user and company information to make a profit. These websites can be accessed just like any other website with just a single click that can make a device or network unusable. An example of this is an attack done on Sony Pictures in 2014 where phishing emails linking to malicious sites that stole login credentials allowing hackers to cause an estimated damage of \$100 million at the time. If proper URL filtering policies were enforced at the time of the attack, those malicious websites would have been blocked preventing the attack. There have been multiple other cases of similar attacks that have caused damage to companies, organizations, and individuals signifying the importance of URL filtering for security purposes. Besides functioning as a safeguard from malicious attacks that can cost companies and people millions of dollars, it also aids in productivity preventing the user from visiting unrelated sites while completing a task. According to Forbes, their research identified that the most blocked categories at companies are gambling and adult sites, with 72% of employers blocking adult sites, and 42% blocking gambling. Forbes additionally stated that 64% of companies reported using firewalls to block unrelated websites from being accessed. These websites are being restricted for good reasons, as Forbes reports that IT professionals estimate that full-time employees are wasting 26 hours annually browsing websites unrelated to work. For modern day businesses, URL filtering and policy implementation are becoming a vital role in beating competition due to both its security and productivity benefits.

Back when URL filtering was first implemented in 1990, they used key word filtering, meaning when a keyword existed within the URL. That site was blocked, preventing the user from accessing the website. This method proved to cause many problems; it resulted in many false positives, over blocking, and needed constant manual updates to keep up with new threats. Because of these problems, modern firewalls implement URL filtering policies by contacting massive servers with URLs organized in categories allowing for more accurate and less resource intensive URL filtering to take place. This is Category Based URL filtering, by having a centralized server with millions of URL organized in categories, lists of malicious websites, and constantly updated lists of new threats modern firewalls can protect companies, users, and customer

information both efficiently and effectively. Additionally, each firewall does not need to be configured to have a database of URLs and what categories they fall in, enabling consistent updates to be pushed to a central server rather than to all firewalls allowing easier maintenance and management.

Before the advent of HTTPS, HTTP was used when visiting a website which is a less secure way of access, but because the traffic using the HTTP protocol was not encrypted it allowed for firewalls to easily read the traffic and block the URL based off what it was categorized as and if it matched with the list of blocked or allowed URLs. Due to the unsecure nature of the HTTP protocol, a newer and more secure HTTPS or Hyper Text Transfer Protocol Secure was created. With the creation of the HTTPS protocol, it made a challenge for firewalls to see what sites a user was accessing as all traffic carried by the HTTPS protocol was encrypted. To effectively and accurately implement URL filtering with HTTPS firewalls had two options: The first doesn't need encryption, when a user device establishes an encrypted connection, it needs to perform a TLS handshake which includes something called the SNI which the firewall can inspect for the unencrypted domain name of the website allowing for the firewall to block the site without the need for decryption. The second option is more advanced but allows the firewall to know the full URL, not just the domain name. For example, with the first method you can only see youtube.com, while with the second method you can see the full URL like youtube.com/watch. The second method works by having the firewall intercept the connection between the client and the webpage, but for this to happen the client-side browser needs to trust the firewall. To make the client browser trust the firewall, the firewall presents a Root Certificate Authority (CA) to the browser; to make the browser trust the CA certificate it usually needs to be deployed to the client's machine. This way the firewall can successfully decrypt the full URL allowing for more precise and accurate URL filtering to be implemented in the modern day where almost all websites use HTTPS.

Lab Summary

In this lab, I configured Category Based URL Filtering that would be appropriate to implement in an elementary school environment. We did this first by setting up a CA certificate and a TLS/SSL service profile. Then configure the URL filtering profile and implement that on the internet policy. We also configured the firewall to consistently update to respond to any new threats identified by the issuer Palo Alto, and updates to the newest operating system.

Lab Commands

First check to make sure an Advanced URL Filtering License is present and not expired in the Devices > Licenses:

Advanced URL Filtering

Date Issued September 29, 2023

Date Expires July 06, 2026

Description Palo Alto Networks Advanced URL License

If the Firewall does not have a valid Advanced URL Filtering License, click “Receive License from License Server” to retrieve the License. This is required for URL filtering to work.

License Management

[Retrieve license keys from license server](#)

[Activate feature using authorization code](#)

[Manually upload license key](#)

Next, configure the firewall to download and install dynamic updates in Devices > Dynamic Updates. Then Schedule the firewall to download and install the updates in the shortest time interval.

Applications and Threats Update Schedule



Recurrence

Minutes Past Half-Hour

Action

☐ Disable new apps in content update

Threshold (hours)

A content update must be at least this many hours old for the action to be taken.

Allow Extra Time to Review New App-IDs

Set the amount of time the firewall waits before installing content updates that contain new App-IDs. You can use this wait period to assess and adjust your security policy based on the new App-IDs.

New App-ID Threshold (hours)

Delete Schedule

OK

Cancel

GlobalProtect Clientless VPN Update Schedule



Recurrence

Minutes Past Hour

Action

Delete Schedule

OK

Cancel

GlobalProtect Data File Update Schedule



Recurrence

Minutes Past Hour

Action

Delete Schedule

OK

Cancel

WildFire Update Schedule



Recurrence

Action

☐ Synchronize content with HA peer after download/install

Delete Schedule

OK

Cancel

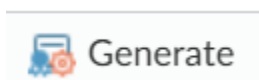
Once the dynamic updates are configured, navigate to Devices > Software to install the latest Operating System.

VERSION	SIZE	SHA256	RELEASE DATE	AVAILABLE	CURRENTLY INSTALLED	ACTION
10.2.16-h4	459 MB	ff075fd6979424f56c3f58dd0ea57279d7382322f619e92d74df07c368d2fb44	2025/09/25 04:27:05			Download
10.2.0	556 MB	dcfa2f15160a29dea26bdf1bcd1b4d20a51d542d7da6fa3c35821783bb54940e	2022/02/27 15:31:56	Downloaded		Validate Export Install
10.1.14-h13	418 MB	4003f28c312151395637a54031a3cd609c6564a472c143ab3eec723f295615ff	2025/04/04 09:34:36			Download
10.1.0	540 MB	6ae2c8f719e8832dec2ebbf42a6c716cda2022dfb19152f8e8694c953737f4	2021/06/02 04:15:33	Downloaded		Validate Export Install
9.1.18	295 MB	b5e79a8d7f7ce2082437408a6f48992e33e9a5209d90420f81daf586b7ae41a	2024/02/27 05:29:41			Download
9.1.17	295 MB	1a96ee7d68d11389aa021af6e9f423a98e33820077a845a90d7325d4c8153d61	2023/12/11 04:34:34			Download
9.1.0	421 MB	74a8883cf58e35ae9da0caf88623653b57e229a583cbe2cea365f8e1d5336574	2019/12/13 08:51:48			Download

Then Download and Install the newest Operating System. If nothing shows up, then click the Check Now button on the bottom left, to load the available Operating System.



To start configuring URL filtering, generate a CA certificate. Do this by navigating to Device > Certificate Management > Certificates and click Generate.



Next Configure the certificate. Name it, find the FDQN by consoling into the firewall and typing the command: “show dns-proxy fqdn all” and enter the shown Fully Qualified Domain Name (FDQN) into the common name. To make sure it is a self-signed CA certificate, you must check the Certificate Authority check box.

Generate Certificate ?

Certificate Type ☒ Local ☐ SCEP

Certificate Name

Common Name
IP or FQDN to appear on the certificate

Signed By

☒ Certificate Authority
 ☐ Block Private Key Export

OCSP Responder

Cryptographic Settings

Algorithm

Number of Bits

Digest

Expiration (days)

Certificate Attributes

☐	TYPE	VALUE

+ Add
- Delete

Generate

Cancel

Configure another certificate signed by the root CA certificate for the common name can be anything.

Generate Certificate?

Certificate Type

☒ Local ☐ SCEP

Certificate Name

Url Filtering

Common Name

Urk Filtering

IP or FQDN to appear on the certificate

Signed By

Root Self-Signed Certificate

☐ Certificate Authority ☐ Block Private Key Export

OCSP Responder

Cryptographic Settings

Algorithm

RSA

Number of Bits

2048

Digest

sha256

Expiration (days)

365

Certificate Attributes

☐	TYPE	VALUE
--------------------------	------	-------

+ Add

- Delete

Generate

Cancel

Next configure the SSL/TLS profile at Device > SSL/TLS profile with the second certificate signed by the CA Certificate. Make sure it accounts for the greatest number of versions.

SSL/TLS Service Profile?

Name

URL_Filter_SSL/TLS_Profile

Certificate

Url Filtering

Protocol Settings

Min Version

TLSv1.0

Max Version

TLSv1.2

OK

Cancel

Next navigate to Objects > URL Filtering and create a new URL filtering profile by clicking “Add” on the bottom left of the screen. Then, configure the appropriate profile for network users. Choose from Override, Block, or Alert do not choose to allow

as alert does the same thing but logs when the policy is being used helping to achieve better security.

URL Filtering Profile

Name

URL filtering rules

Description

Categories

URL Filtering Settings

User Credential Detection

HTTP Header Insertion

Inline Categorization

90 items

☐	CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
☐	abused-drugs	override	allow
☐	adult	override	allow
☐	AI-code-assistant	override	allow
☐	AI-conversational-assistant	override	allow
☐	AI-data-and-workflow-optimizer	override	allow
☐	AI-media-service	override	allow
☐	AI-meeting-assistant	override	allow
☐	AI-platform-service	override	allow

* indicates a custom URL category, + indicates external dynamic list

[Check URL Category](#)

OK

Cancel

Next configure the admin override password. Navigate to Device > Setup > Content-ID and under URL Admin Override click add and configure a password.

URL Admin Override

Password

••••••••

Confirm Password

••••••••

SSL/TLS Service Profile

URL_Filter_SSL/TLS_Profile

Mode

☒ Transparent

☐ Redirect

OK

Cancel

Next configure the override timeout which dictates how long the page will remain open after inputting the override password.

URL Filtering

URL Continue Timeout (min)

15

URL Admin Override Timeout (min)

15

URL Admin Lockout Timeout (min)

30

☐ Hold client request for category lookup

Category lookup timeout (sec)

2

PAN-DB Server

Cloud list separated by commas

☒ Append Ending Token

OK

Cancel

Finally, navigate to policies and configure your internet rules by going to Policies > Security > Internet Policy > Action and specify the URL profile.

Security Policy Rule ?

General | Source | Destination | Application | Service/URL Category | **Actions** | Usage

Action Setting

Action: Allow

☐ Send ICMP Unreachable

Profile Setting

Profile Type: Profiles

Antivirus: default

Vulnerability Protection: default

Anti-Spyware: default

URL Filtering: URL filtering rules

File Blocking: None

Data Filtering: None

WildFire Analysis: default

Log Setting

☐ Log at Session Start

☒ Log at Session End

Log Forwarding: None

Other Settings

Schedule: None

QoS Marking: None

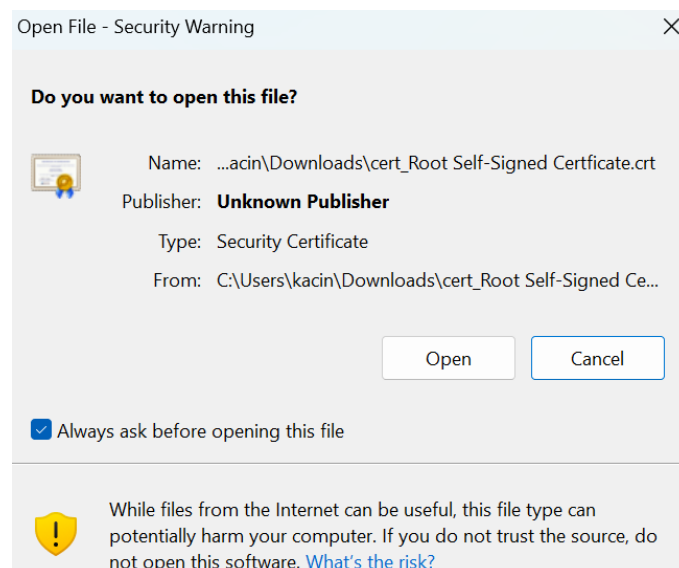
☐ Disable Server Response Inspection

OK Cancel

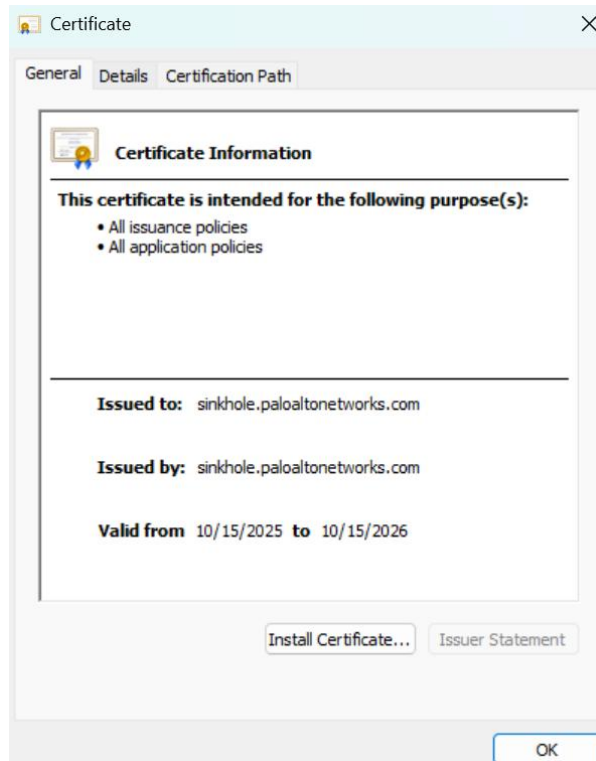
These basic configurations should be enough for basic blocking URL filtering, but it cannot inject an override page as a decryption policy has not been set up yet for HTTPS. To configure HTTPS filtering, the device needs to trust the root CA certificate so that the firewall can establish a connection between the webserver so that it can decrypt the HTTPS traffic.

To make your local machine trust the root CA certificate go to Devices > Certificates, select the CA certificate and click “Export”. Double click the download and follow the instructions.

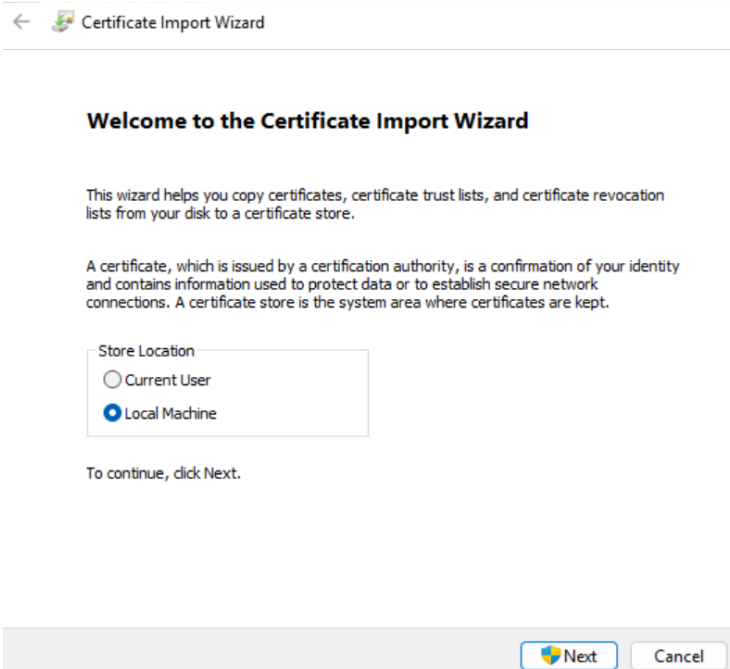
Click Open



Install the Certificate



Select store in local machine and click next, you will need admin permissions



Manually choose where to store the certificate, and then click browse

Certificate Store

Certificate stores are system areas where certificates are kept.

Windows can automatically select a certificate store, or you can specify a location for the certificate.

☐ Automatically select the certificate store based on the type of certificate

☒ Place all certificates in the following store

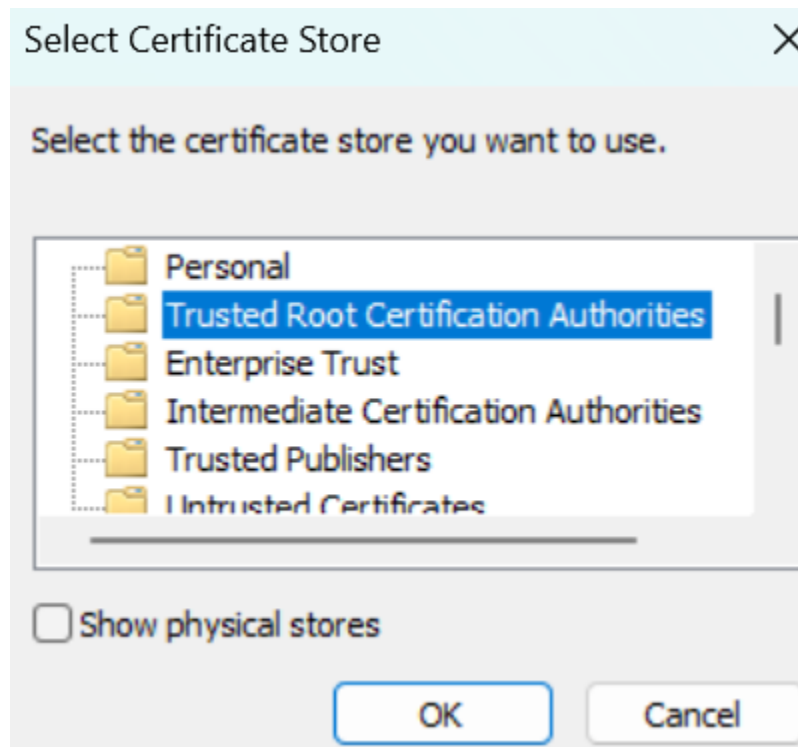
Certificate store:

Browse...

Next

Cancel

Select “Trusted Root Certificate Authorities” and click OK.



Click Finish, and now the local machine trusts the firewall.

Next, to decrypt the HTTPS traffic, the firewall needs a decryption policy. Navigate to Policies > Decryption and click add to create a new policy. Create a name and select the inside source zone and outside source zone.

Decryption Policy Rule

General

Source

Destination

Service/URL Category

Options

☐ Any	☒ Any	any	any
☐ SOURCE ZONE ^	☐ SOURCE ADDRESS ^	☐ SOURCE USER ^	☐ SOURCE DEVICE ^
☐ Trust-L3			
+ Add - Delete	+ Add - Delete	+ Add - Delete	+ Add - Delete

☐ Negate

OK

Cancel

Decryption Policy Rule

General

Source

Destination

Service/URL Category

Options

☐ Any	☒ Any	☒ Any
☐ DESTINATION ZONE ^	☐ DESTINATION ADDRESS ^	☐ DESTINATION DEVICE ^
☐ Untrust-L3		
+ Add - Delete	+ Add - Delete	+ Add - Delete

☐ Negate

OK

Cancel

Select HTTP and HTTPS services in the Service/URL Category tab.

Decryption Policy Rule ?

General
Source
Destination
Service/URL Category
Options

select

SERVICE ^

☐ service-http
☒ service-https

URL CATEGORY ^

☒ Any

+ Add

- Delete

+ Add

- Delete

OK

Cancel

Then, in the Options tab select the action to decrypt, and the type as SSL Forward Proxy, and Profile as default.

Decryption Policy Rule ?

General
Source
Destination
Service/URL Category
Options

Action

☐ No Decrypt
☒ Decrypt

Type

SSL Forward Proxy

Decryption Profile

default

Log Settings

☒ Log Successful SSL Handshake
☒ Log Unsuccessful SSL Handshake

Log Forwarding

None

OK

Cancel

To then verify that the URL filtering works go to a website that would need an override password to enter like <https://walmart.com> and an override page should be injected.



Web Page Blocked

The web page you are trying to visit has been blocked in accordance with company policy. Please contact your system administrator if you believe this is an error.

User: 192.168.1.2

URL: https://walmart.com/

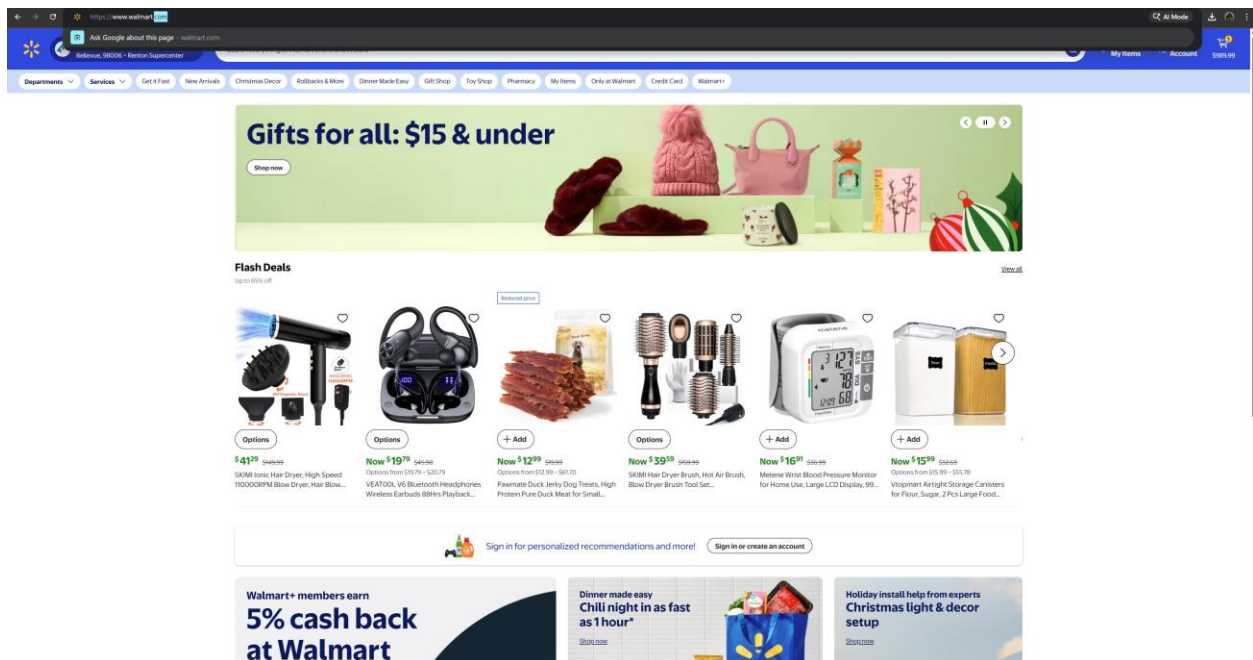
Category: shopping

If you require access to this page, have an administrator enter the override password here:

Continue

[Return to previous page](#)

Enter the password, and access to the webpage should be allowed.



Configuration of URL filtering is complete.

Problems

During this lab, a major problem I first had was that I didn't configure anything related to certificates which made it, so my URL filtering did not work when it came to HTTPS filtering. But once I configured the proper certificates and the TLS and SSL service profiles, it still did not work. This was because I didn't fully understand how URL filtering works when it comes to HTTPS, and that just because I made a certificate does not mean my local device or browser trusted the certificate I made on the firewall. Once I had realized this, I was able to learn how to install a certificate onto my local device so that it would trust my certificate so that my firewall could establish a connection between my computer and the webserver as the man in the middle. I then had to learn that I needed to manually implement a decryption policy on the firewall because the HTTPS traffic is encrypted and to actually do URL filtering the firewall would need to decrypt the HTTPS traffic to see the full URL so that it can match it against the URL categories I set to override. These experiences helped me to understand what URL filtering is, how it works, and how to configure it.

Conclusion

This lab has helped me understand exactly how a firewall preforms URL filtering by being the man in the middle that the computer can trust. It has also taught me how HTTP and HTTPS protocols work and how HTTPS configurations need extra steps for implementing URL filtering. Understanding these concepts has built a strong foundation in configuring, managing, and maintaining a network with firewalls implementing policies on a network.